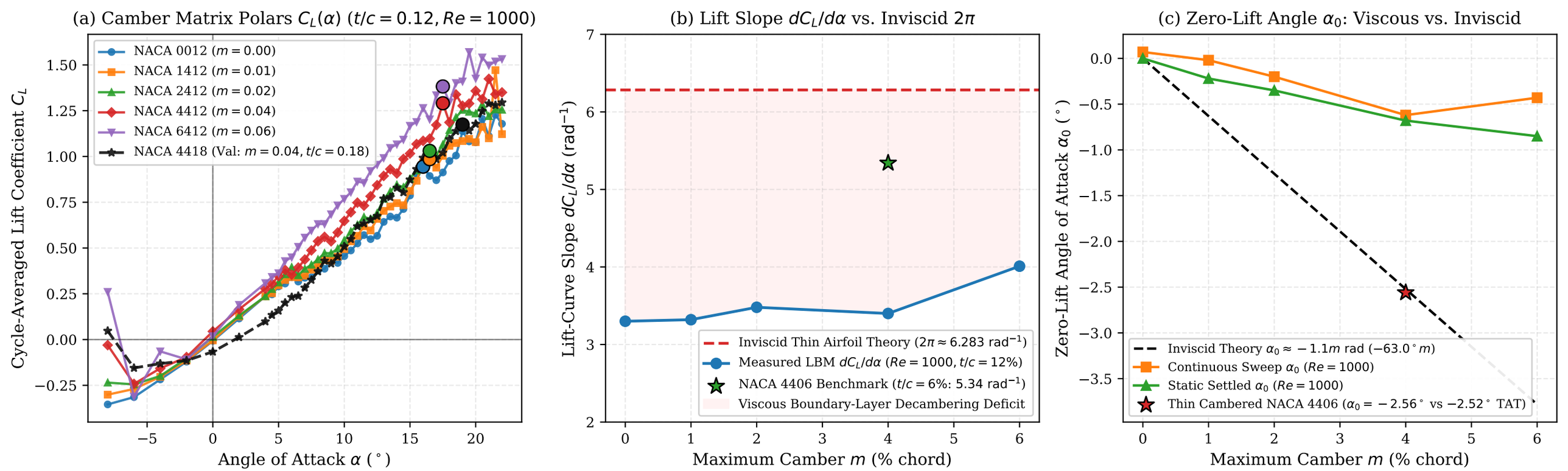
(a) Camber Matrix Polars $C_L(\alpha)$ ($t/c = 0.12, Re = 1000$)
Cycle-Averaged Lift Coefficient C_L
Angle of Attack α (°)
NACA 0012 ($m = 0.00$)
NACA 1412 ($m = 0.01$)
NACA 2412 ($m = 0.02$)
NACA 4412 ($m = 0.04$)
NACA 6412 ($m = 0.06$)
NACA 4418 (Val: $m = 0.04$, $t/c = 0.18$)
(b) Lift Slope $dC_L/d\alpha$ vs. Inviscid 2π
Lift-Curve Slope $dC_L/d\alpha$ (rad^{-1})
Maximum Camber m (% chord)
Inviscid Thin Airfoil Theory ($2\pi \approx 6.283$ rad^{-1})
Measured LBM $dC_L/d\alpha$ ($Re = 1000$, $t/c = 12\%$)
NACA 4406 Benchmark ($t/c = 6\%$: 5.34 rad^{-1})
Viscous Boundary-Layer Decambering Deficit
(c) Zero-Lift Angle α_0: Viscous vs. Inviscid
Zero-Lift Angle of Attack α_0 (°)
Maximum Camber m (% chord)
Inviscid Theory $\alpha_0 \approx -1.1m$ rad ($-63.0° m$)
Continuous Sweep α_0 ($Re = 1000$)
Static Settled α_0 ($Re = 1000$)
Thin Cambered NACA 4406 ($\alpha_0 = -2.56°$ vs $-2.52°$ TAT)